

HW 2

1. What is the largest $p \in \mathbb{N}$ such that all integers in the interval $[-p, p]$ are exactly representable in IEEE double-precision arithmetic? For single-precision arithmetic?

largest natural num: $0; \underbrace{00\dots 0110100}_{11 \text{ bits}}; \underbrace{111\dots 1}_{52 \text{ bits}}$
 $32+16+4$

$$(1.111\dots 1)_2 \times 2^{52}$$

An integer needs $\log_2 |n| + 1$ bits to represent in binary. Since we are only concerned with the largest natural number p in the range $[-p, p]$ that can be represented exactly, the bit representation must fit within the 52 bits of the significand and the implicit leading bit of one. Thus, exact representability requires

$$\log_2 |n| + 1 \leq 53 \quad \text{or} \quad |n| \leq 2^{53}$$

Testing the boundaries of this, 2^{53} can be represented exactly as one followed by 53 zeroes in binary. Thus the largest $p \in \mathbb{N}$ such that all integers are represented exactly is 2^{53} , or all integers in the range $[-2^{53}, 2^{53}]$. For single-precision p would be 2^{24} since there are 23 significand bits and one leading bit.

2. Stochastic rounding $R = \text{round}(x)$ is a RV

$$P(R = \lceil x \rceil) = \frac{x - \lfloor x \rfloor}{\lceil x \rceil - \lfloor x \rfloor}$$

$$P(R = \lfloor x \rfloor) = \frac{\lceil x \rceil - x}{\lceil x \rceil - \lfloor x \rfloor}$$

$$E(R) = \lceil x \rceil \cdot P(R = \lceil x \rceil) + \lfloor x \rfloor \cdot P(R = \lfloor x \rfloor)$$

$$= \lceil x \rceil \cdot \frac{x - \lfloor x \rfloor}{\lceil x \rceil - \lfloor x \rfloor} + \lfloor x \rfloor \cdot \frac{\lceil x \rceil - x}{\lceil x \rceil - \lfloor x \rfloor}$$

$$= \frac{\lceil x \rceil x - \lceil x \rceil \lfloor x \rfloor + \lfloor x \rfloor \lceil x \rceil - \lfloor x \rfloor x}{\lceil x \rceil - \lfloor x \rfloor}$$

$$= \frac{\lceil x \rceil x - \lfloor x \rfloor x}{\lceil x \rceil - \lfloor x \rfloor} = \frac{x(\lceil x \rceil - \lfloor x \rfloor)}{\lceil x \rceil - \lfloor x \rfloor} \rightarrow \lceil x \rceil - \lfloor x \rfloor = 1$$

$$= \boxed{x}$$

3. Let $\mathcal{P}_n$ be the vector space of polynomials w/ degree at most n .

a) Basis for $\mathcal{P}_n$

$$\left\{ \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \end{bmatrix}}_{n+1 \text{ vectors}} \right\} \text{ or } \{1, x, x^2, \dots, x^n\}$$

The dimension of this space is $\boxed{n+1}$

b) Derivative operator $v, w \in \mathcal{P}_n \quad c \in \mathbb{R}$

$$T: \mathcal{P}_n \rightarrow \mathcal{P}_{n-1}$$

$$\oplus \quad T(v+w) = (v+w)' = v' + w' = T(v) + T(w) \quad \checkmark$$

$$\odot \quad T(cv) = (cv)' = cp' = cT(v) \quad \checkmark$$

$\therefore$ the derivative operator is a linear transformation

$$T(1) = 0 \quad T(x) = 1 \quad T(x^2) = 2x \quad T(x^n) = nx^{n-1}$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 2 & 0 & \dots & 0 \\ 0 & 0 & 0 & 3 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & n \end{bmatrix} \text{ in } \mathcal{P}_{n-1} \text{ vector space}$$

c) Subset $W \subseteq \mathcal{P}_n$

The subset of polynomials with zero constant term

$$W = \text{span}\{x, x^2, \dots, x^n\}$$

① Contains 0? Yes, $p(x) = 0$ satisfies $p(0) = 0$

② addition: $p, q \in W, p+q \in W$

$$\text{if } p(0) = 0 \text{ and } q(0) = 0, \text{ then } (p+q)(0) = 0$$

③ scalar mult: $p \in W, c \in \mathbb{R}$

$$\text{if } p(0) = 0, \text{ then } (cp)(0) = cp(0) = c \cdot 0 = 0$$

$\therefore W$ is a subspace

4. Let $v_1, \dots, v_n$ be a basis for vector space V . Suppose $w \in V$, where, by definition of basis, can be represented as

$$w = a_1 v_1 + a_2 v_2 + \dots + a_n v_n, \quad a_1, \dots, a_n \in \mathbb{R}$$

$$w = \beta_1 v_1 + \beta_2 v_2 + \dots + \beta_n v_n, \quad \beta_1, \dots, \beta_n \in \mathbb{R}$$

Here, we have 2 linear combinations of w , and if we subtract them,

$$w - w = a_1 v_1 + \dots + a_n v_n - (\beta_1 v_1 + \dots + \beta_n v_n) = 0$$

$$= (a_1 - \beta_1) v_1 + \dots + (a_n - \beta_n) v_n = 0$$

Since $\{v_1, \dots, v_n\}$ is linearly independent, the only solution to this equation is if $(a_1 - \beta_1) = \dots = (a_n - \beta_n) = 0$, which proves uniqueness.

5. Basis for null space of A

$$A = \begin{bmatrix} 1 & 0 & -2 & -1 \\ 1 & 1 & -2 & 0 \\ 2 & 0 & -4 & -2 \\ 0 & 1 & 0 & 1 \end{bmatrix} \xrightarrow[R2 = -2R1 + R2]{R2 \leftrightarrow R4} \begin{bmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & -2 & 0 \end{bmatrix} \xrightarrow[R3 \rightarrow -R1 + R3]{R3 \leftrightarrow R4} \begin{bmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Ax = 0 = \begin{cases} x_1 - 2x_3 - x_4 = 0 \\ x_2 + x_4 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = 2x_3 + x_4 \\ x_2 = -x_4 \end{cases} = x_3 \begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

Basis for null space of A is $\left\{ \begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}$ $\leftarrow \dim(\text{null}(A)) = 2$

6. Programming Question

a) in code below

b) Total flops

for each C_{ij} , there are n multiplication ops from $A_{ik}B_{kj}$

$n-1$ addition ops when adding the product to the sum

following, per C_{ij} : $n + (n-1) = 2n - 1$

Since there are n^2 C_{ij} 's, $O(n) = n^2(2n-1) = 2n^3 - n^2$

$$\Rightarrow \boxed{O(n^3)}$$

c) in code below

some matrix entries exceed the bound.

```

function C = multiplyMatrices(A, B)
%HW2 Floating-point matrix multiplication
% This function takes in 2 matrices A and B and outputs their n x n
% product C = AB
n = size(A, 1);
C = zeros(n,n);

    for i = 1:n
        for j = 1:n
            for k = 1:n
                C(i, j) = C(i, j) + A(i, k) * B(k, j);
            end
        end
    end
end

A1 = [1 2 3;
      4 5 6;
      7 8 9];
B1 = [9 8 7;
      6 5 4;
      3 2 1];
C1 = multiplyMatrices(A1, B1);
disp('Resulting Matrix C (part a):');
disp(C1);

A = [ 1/sqrt(5)*ones(1,5);
      1/sqrt(2) 0 -1/sqrt(2) 0 0;
      -1/2 1/2 -1/2 1/2 0;
      0 1/sqrt(2) 0 -1/sqrt(2) 0;
      1/(2*sqrt(5))*[1 1 1 1 -2] ];
B = A';
C = multiplyMatrices(A, B);
disp('Resulting Matrix C (part c):');
disp(C);

C_exact = eye(size(A,1));
E = abs(C - C_exact);

n = size(A,2); % length of inner product
gamma_n = n*eps/(1 - n*eps);

% Compute sum of absolute products for each entry
entrywise_bound = zeros(size(A,1));
for i = 1:size(A,1)
    for j = 1:size(A,1)
        entrywise_bound(i,j) = gamma_n * sum(abs(A(i,:)) .* abs(B(:,j)'));
    end
end

% Check that error <= bound for all entries
check = E <= entrywise_bound;

```

```

if all(check, 'all')
    disp('All entries satisfy the a priori error bound.');
```

else

```

    disp('Some entries exceed the bound.');
```

end

Resulting Matrix C (part a):

30	24	18
84	69	54
138	114	90

Resulting Matrix C (part c):

1.0000	0	0	0	0.2000
0	1.0000	0	0	0
0	0	1.0000	0	0
0	0	0	1.0000	0
0.2000	0	0	0	0.4000

Some entries exceed the bound.

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